

AN-01: Architecting a Direct, 3-phase Energy Meter with Shunts using V9240

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REVISION HISTORY

Date	Version	Description	
2018-10-10	V1.0	Initial Release	

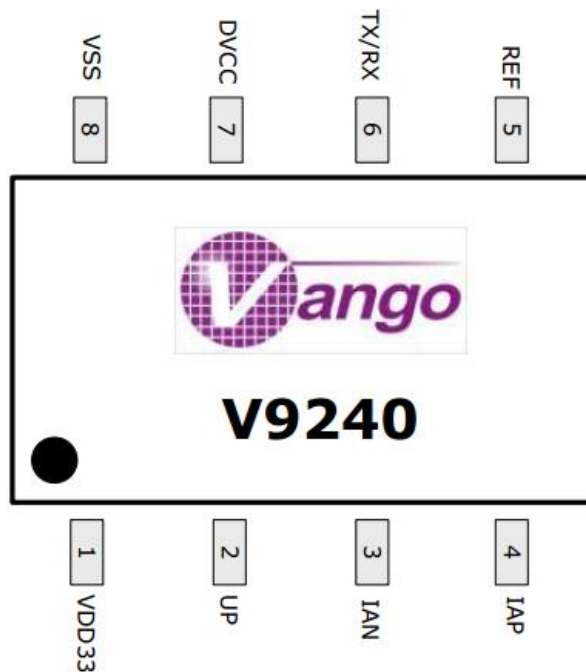
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1. GENERAL DESCRIPTION

A 3 shunt sensing poly-phase meter designed with 3 single phase AFE V9240s and a MCU is proposed in this document.

V9240 is a multifunction, high accuracy, ultralow power, single-phase power measurement IC with baud rate adaption UART serial interface.

V9240 Pin Definations



2. SCHEMATIC DESIGN

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V9240 includes two channels: one current channel and one voltage channel. Each V9240 measures one phase of 3-phase meter including current RMS, voltage RMS, power, frequency, phase etc.

2.1. SYSTEM DESIGN

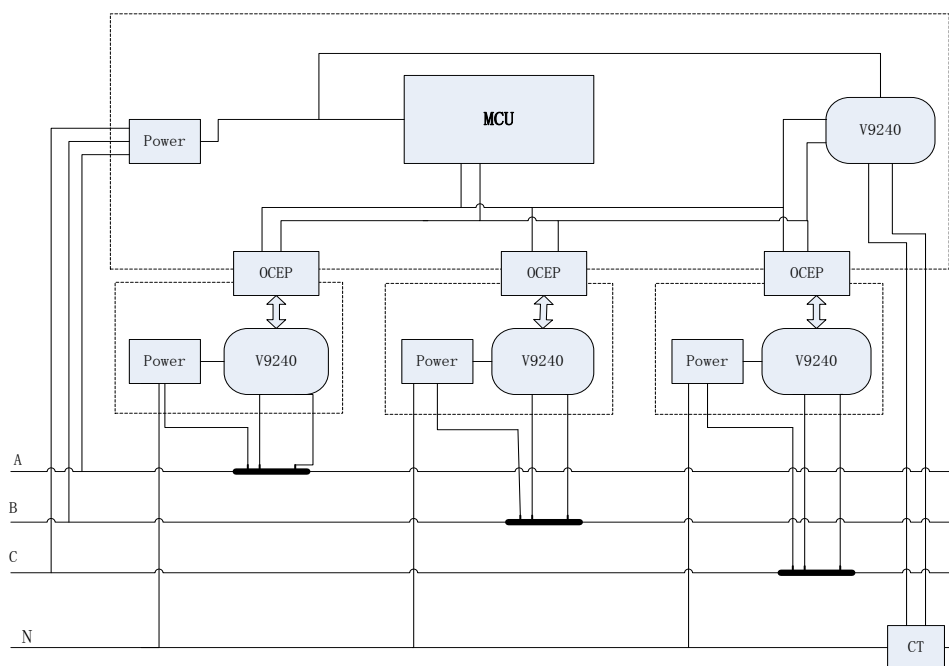


Figure 2-1 Three-phase four-wire meter system block diagram

Figure 2-1 shows the architecture of a meter designed with 2 V9240. Each V9240 corresponds with one phase. The reference terminal of the shunt becomes the ground of the isolated side of the V9240. The neutral line determines the ground of MCU that manages the V9240 devices. The MCU communicates with V9240 via UART. The power supplies of phase A, phase B and phase C are isolated.

2.2. SAMPLING CIRCUITS

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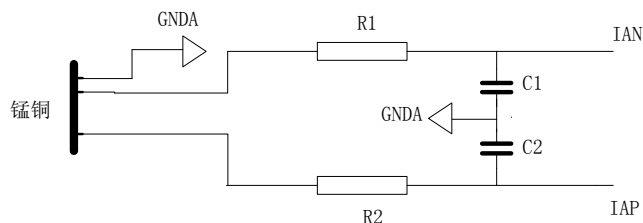


Figure 2-2: Current Sampling Circuits

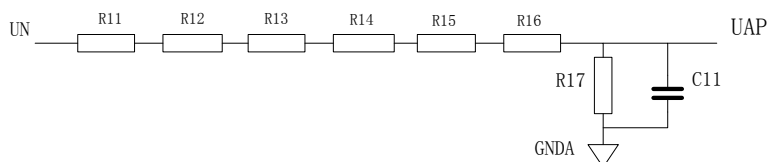


Figure 2-3: Voltage Sampling Circuits

For the current channel, shunt resistor can be used for analog inputs. For voltage channel, a resistor-divider network can be used for analog inputs.

There are some recommendations as below:

1. Each input has a maximum voltage of $\pm 200\text{mV}$, and each pair of a maximum differential voltage of $\pm 400\text{mV}$. The value of input signal multiplying by PGA gain (both analog and digital) must be no more than the voltage reference voltage (1.1V). We recommend that current signal at about $1\sim 2\text{mV}$, analog PGA gain set to 32, digital PGA gain set to 1, voltage signal at about 65mV , analog PGA gain set to 4, digital PGA gain set to 1.
2. Antialiasing filter between the IAP and IAN pin is needed, where R and C values are $360\ \Omega$ and 33nF .
3. The phase A, B, or C to neutral voltages are sensed with the voltage dividers R1~R16. The resistor divider network is composed of six $200\text{k}\Omega$ resistors, and R17 is equal to $360\ \Omega$, C11 is equal to 33nF .

2.3. INTERFACE CIRCUIT

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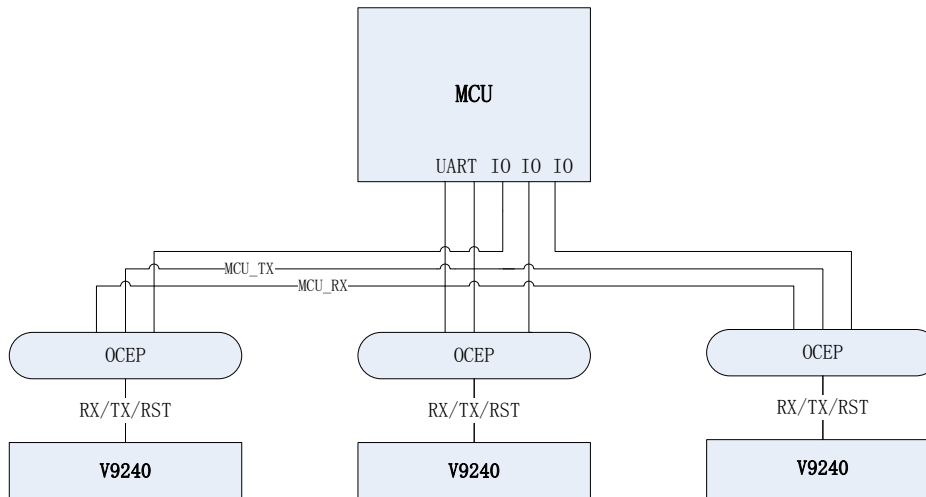


Figure 2-4: Connections between three V9240s and the MCU

The V9240 integrates a baud rate adaption UART interface, and the datasheet provides extensive details on the protocol used to read and write the device registers.

Suppose that the MCU is connected to V9240s as showed in Figure 2-4. The UART port of V9240 is only one pin "RX" which has three kinds of fuction including transmission, receiving and resetting the chip. Holding logic low for at least 64ms can reset V9240.

There is no chip selection pin in V9240, so additional three input/ouput(I/O) lines of the MCU are required to control each OCEP to accomplish the communication between the MCU and each V9240.

批注 [门长有1]: ?

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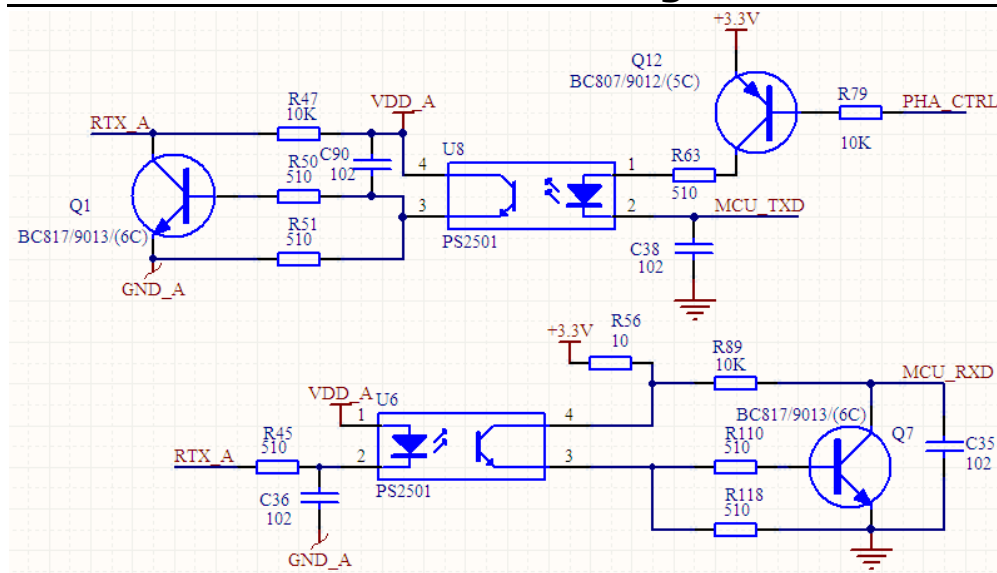


Figure 2-5: UART Interface Circuit

Figure 2-5 shows the UART interface circuit between V9240 and MCU

1. The resistors to drive the OCEPs are very important which ensure the reliable communication over the -40°C to $+85^{\circ}\text{C}$. Both the value of R63&R45 are recommended to set to 510Ω .
2. The input/output(I/O) line of the MCU, PHA_CTRL, determines to communicate with V9240 by switching the OCEP.
3. The triodes Q1&Q7 provide a better waveform.

批注 [门长有2]: 什么意思?

2.4. HARDWARE NOTICE

1. A $\geq 4.7\mu\text{F}$ and $0.1\mu\text{F}$ decoupling capacitor is needed for DVCC pin.
2. A $1\mu\text{F}$ decoupling capacitor is needed for REF pin.

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3. LAYOUT GUIDELINES

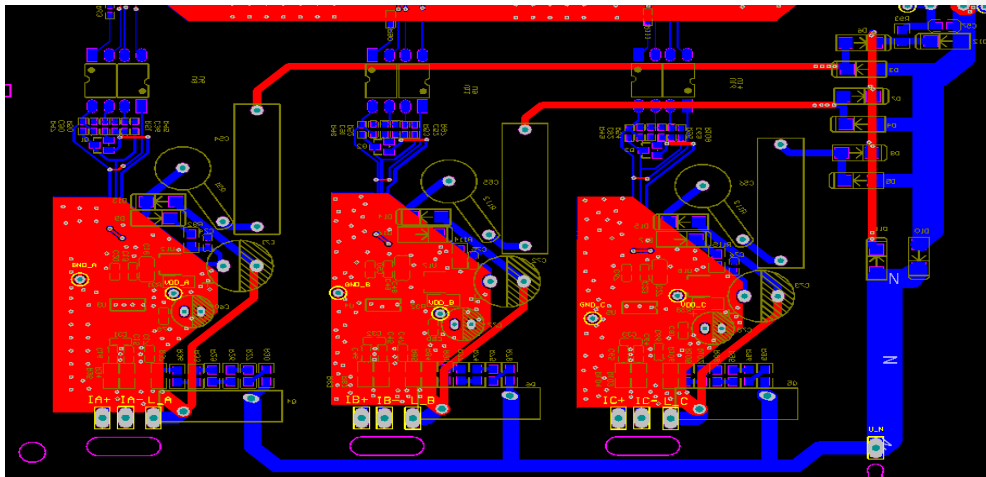


Figure 3-1: Top Layer Layout of the Direct, 3-phase Meter

1. The three V9240s must be isolated.
2. For good EMC performance and high accuracy, big and unbroken area of copper pouring is very necessary.
3. The antialiasing filters, $R/C(360\Omega/33nF)$ should be close to the pins, and the route from PAD to IAP/IAN should be as short as possible.
4. Place the decoupling capacitors between DVCC and GND(10uF and 100nF), REF and GND(1uF) close to the chip.

4. SOFTWARE DESIGN

4.1. ENERGY METERING

V9240 does not provide energy data, and only provides instantaneous power value (160ms update once) and average power value (640ms update once). In consideration of

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high accuracy and reliable communication, we recommend calculating energy by accumulating average power value.

The procedure of calculating energy is composed of three steps, reading power value, accumulating power value and outputting energy pulse.

Reading power value

The MCU reads average power value of each V9240 as power value of each phase, and selects the absolute value or the sum of algebra to measure total power value.

Accumulating power value

We should create a 32-bit variable to accumulate total power value at the frequency of n/s, and compare with the threshold value. If the variable exceeds the threshold value, it should be cleared.

Outputting energy pulse (CF)

We define that CF outputs once when the variable exceeds the threshold twice.

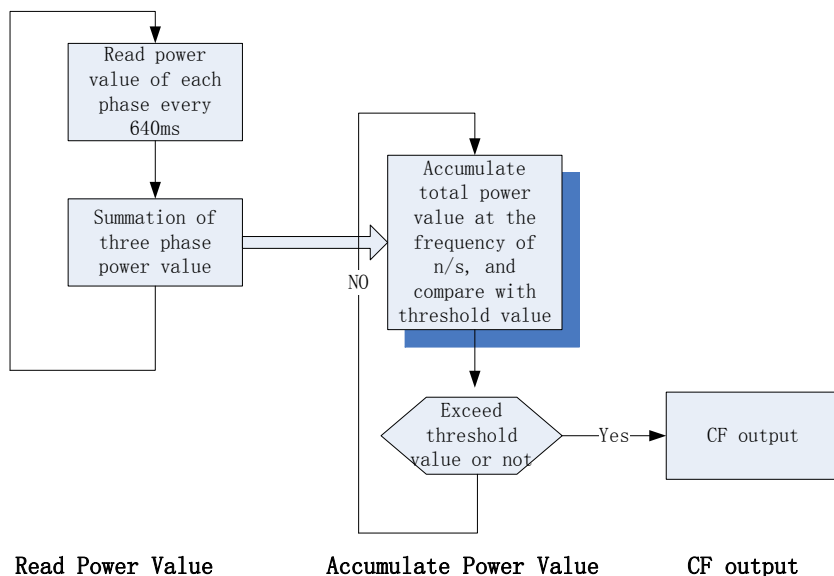


Figure 4-1: Energy metering block diagram

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1. It is much more reliable that reading average power value than instantaneous.
2. Suggest that reading power value every 640ms, and accumulating power value at a frequency of 5000/s.
3. Timer should be in auto-reload mode, and the priority of timer interrupt should be configured highest.
4. Recommend that the threshold value is half of standard value to realize 50% duty ratio when the interval time of CF output is less than 80ms.

4.2. SOFTWARE NOTICE

1. Automatic baud rate adaption: support 1200bps~9600bps, and recommend 4800bps.
2. A 11-bit data byte, composed of 1-bit start bit, 8-bit data bits, 1-bit parity(**odd**), and 1-bit stop bit.
3. Hold low logic of RX/TX pin for at least 64ms to reset the chip.
4. Wait 500ms to get correct power value after enabling ADC channel.
5. Another application note provides extensive details about how to get the value of phase angle if necessary.

4.3. V9240 REGISTER CONFIGURATION

Register	Address	Recommended Value	Remarks
SysCtrl	0x0180	0x28000000	Enable voltage(U), current(IA) channel ADC, U channel ADC analog gain X4, IA channel ADC analog gain X32, Bit[18:14] should be configured according to the calculated result to obtain the best metering and temperature performance.
BPFPARA	0x0107	0x806764B6	In order to obtain accurate frequency measurement and phase measurement.
AnaCtrl1	0x0183	0x80000000	In order to obtain accurate frequency measurement and phase measurement.
Configure the three register above, V9240 can run normally.			
PAC	0x00F6		To set gain calibration of active power for A

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			channel
PHC	0x00F7		To set phase calibration of active power
PADCC	0x00F8		Low-current signal calibration of active power for A channel
QAC	0x00F9		To set gain calibration of reactive power for A channel
QADCC	0x00FB		Low-current signal calibration of reactive power for A channel
IAC	0x00FD		A channel current RMS gain calibration
UAC	0x00FF		U channel voltage RMS gain calibration
The meter calibration related registers should be configured according to the calculated result, detail calculation formulas according to datasheet.			
CKSUM	0x0109		System control register(0x0180)+calibration registers+checksum register(0x0109) = 0xFFFFFFFF
To set CKSUM register can monitor that whether or not configuration of the registers being changed accidentally, to ensure the system is reliable.			